Evaluation of Automated Clinical Notes for Anesthesia Consultations

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Abstract

This paper presents a novel evaluation framework for assessing AI-generated clinical notes derived from obstetric anesthesia consultations. The system addresses the labor-intensive process where anesthesiologists manually convert patient consultations into structured reports, which diverts valuable clinical time from patient care. Our approach combines automated concept labeling, sentence-by-sentence mapping, and retrieval-augmented generation to detect hallucinations, knowledge gaps, and other documentation issues. The framework delivers structured evaluations containing color-coded highlighted notes, comprehensive scoring rubrics, categorized actionable items, and synthesized assessments—transforming evaluation from a passive, often-skipped assessment into an active clinical review process. Through testing on simulated consultations transcribed via Whisper and AI-generated clinical notes, our findings demonstrate the system effectively approximates physician assessments while significantly accelerating review time. This work contributes a clinically-relevant methodology for evaluating AI-generated content in domains where factual accuracy and completeness have life-critical implications, establishing a foundation for responsibly integrating ambient clinical documentation technologies into healthcare settings.

1 Introduction

In modern healthcare delivery, anesthesiologists face mounting documentation burdens that divert valuable time away from patient care [Gaffney et al., 2022]. Before surgical procedures or deliveries, these specialists conduct comprehensive consultations to assess patient history, evaluate anesthesia risks, and ensure informed consent. Currently, clinicians must manually convert these lengthy, information-dense conversations into structured clinical documentation, a process that can consume hours of clinical time per day. This documentation challenge is particularly acute in obstetric anesthesia, where risk assessments directly impact maternal and neonatal outcomes, and complete, accurate notes are essential for patient safety and legal compliance.

The emergence of ambient clinical documentation technologies, often called "ambient scribes" or "digital scribes," offers a promising solution to this challenge [Tierney et al., 2024]. These AI-powered systems leverage speech recognition and natural language processing to generate clinical notes directly from doctor-patient conversations, potentially eliminating hours of manual documentation work. However, there is a secondary challenge in the deployment of such systems in ensuring these AI-generated notes accurately reflect the consultation content while meeting clinical documentation standards. Shah et al. demonstrate that while LLMs have impressive capabilities in generating

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human-like text, their tendency toward hallucination and incomplete knowledge representation poses significant risks in clinical settings [Shah, 2024].

We describe a specialized evaluation framework for AI-generated obstetric anesthesia notes that serves a dual purpose; it evaluates note quality and also functions as a second layer of validation for clinicians reviewing ambient-scribed documentation. Given practical constraints, clinicians often struggle with comprehensively checking LLM-generated notes for accuracy, assuming that the notes are correct. By combining physician reviews, LLM-based evaluation, and retrieval-augmented generation, our system analyzes simulated conversations transcribed via Whisper and LLM-generated notes. This structured approach not only identifies problematic content, but presents findings in ways that significantly reduce the cognitive burden of note review-a critical consideration given that clinicians must ultimately verify and take responsibility for all documentation.

Our findings demonstrate that this framework effectively approximates physician assessments while presenting results in a standardized format that significantly accelerates clinical review time-a critical factor for practical implementation in anesthesia workflows. By serving as both quality assessment and intelligent review assistant, our system addresses the complete clinical documentation lifecycle from generation to validation.

Contributions:

This research makes several contributions to the field of medical AI and clinical documentation evaluation. This work introduces a novel concept-based evaluation framework that bridges the gap between traditional NLP metrics and clinical usefulness by mapping semantic elements between transcripts and AI-generated notes at a sentence level. Unlike existing approaches that rely on generic text similarity measures such as those described by Singhal et al., our framework captures the nuanced requirements of clinical documentation where paraphrasing is common but factual consistency is critical [Singhal et al., 2023]. By integrating retrieval-augmented generation (RAG) with structured LLM-based evaluation, we demonstrate how emerging generative AI technologies can be deployed not just for content creation but for context-aware clinical quality assessment. Our structured reporting methodology-featuring color-coded concept highlighting, standardized issue taxonomy, and prioritized actionable recommendations-provides a blueprint for designing AI-assisted review systems that reduce cognitive load for clinicians while maintaining rigorous quality standards. This work establishes a foundation for responsible integration of ambient clinical documentation technologies in healthcare settings while providing generalizable principles for evaluating AI-generated content in domains where factual accuracy and completeness have life-critical implications.

2 Related Works

The evaluation of AI-generated clinical documentation is critical as these systems increasingly influence clinical decision-making and patient care [Meng et al., 2024]. While several approaches exist for generating clinical documentation with AI, evaluation frameworks remain less standardized. Traditional evaluation methods for LLMs such as the BLEU and ROUGE scores fail in the medical context because they primarily measure lexical overlap without considering clinical accuracy and relevance [Singhal et al., 2023]. To address these limitations, researchers have proposed alternative evaluation approaches that better reflect clinical relevance. One notable direction involves the use of embedding-based similarity metrics, which move beyond lexical overlap into semantic similarity. Among these, BERTScore by Zhang et al. [2020] has emerged as a widely adopted metric. BERTScore leverages contextual embeddings from pretrained language models like BERT to compute token-level similarity between candidate and reference texts. This allows it to capture paraphrases, contexts, and subtle variations in clinical language that traditional metrics often fail in. However, while BERTScore and other embedding-based similarity metrics are an advancement over traditional metrics, they are significantly limited in the medical domain due to the nuanced nature of clinical language. Most medical concepts are in semantic proximity in the embedding space which can be challenging and may lead to potentially dangerous misinterpretations. For instance, our experiments revealed that embeddings failed to distinguish between clinically distinct concepts such as "multiple pregnancy" (twins/triplets) and "this patient is pregnant with twins," treating them as highly correlated despite their fundamental clinical difference (Figure 1).

	Sentence	Gemini	Ref_Raw_BERT	Ref_BioBERT_SBERT
0	History of Present Illness:	0.05	0.549700	0.046255
1	Jane Blank, a 35-year-old female at 33w3d gest...	0.15	0.904963	0.518825
2	The patient is pregnant with twins.	0.75	0.738074	0.636033
3	Anesthesia Approach: Likely combined spinal-ep...	0.15	0.749565	0.492247

Figure 1: Direct comparison between Gemini (LLM) vs BERT (embedding) vs Bio-BERT (embedding), noting that embedding similarity does not accurately identify sentences relating to Multiple Gestation

To address the shortcomings of both semantic and quantitative text similarity metrics, recent studies have developed evaluation frameworks that incorporate domain expertise. For example, Seo et al. developed a human-in-the-loop framework where clinicians evaluated LLM-generated emergency department notes across multiple dimensions including clinical accuracy, structure, and conciseness [Seo et al., 2024]. Their methodology involves a manual annotation of specific error types to characterize the nature of inaccuracies. This approach establishes a useful benchmark for clinical acceptability, though it depends primarily on physician assessment and does not incorporate automated tools for detecting content misalignment or inconsistencies at the sentence level.

Other studies have explored the use of LLMs as evaluators themselves or what is now termed LLM-as-a-judge [Zheng et al., 2023]. Wilhelm et al. (2023) compared physician evaluations of treatment recommendations with those generated by GPT-4 using a modified DISCERN instrument (mDISCERN), a validated tool for assessing the quality of health information [Wilhelm et al., 2023]. Their study demonstrated a significant correlation between human expert and LLM evaluations across multiple medical specialties, suggesting that well-prompted LLMs could serve as preliminary screeners for medical content quality. However, this approach still requires validation against expert judgment and faces challenges in consistently identifying potentially harmful information.

Chen et al. (2025) provides a thorough overview of current strategies for evaluating large language models in healthcare and emphasize the importance of using a mix of automated metrics and human judgment across a wide range of clinical tasks [Chen et al., 2025]. They stress the importance of human-centered and context-sensitive tools in medical scenarios. Our work builds on this insight by introducing a concept alignment framework designed to help obstetric anesthesiologists quickly spot hallucinations and errors in AI-generated clinical notes.

3 Method

3.1 Problem Definition

Clinical note taking and interpretation present an overwhelming time burden for specialized obstetric anesthesiologists. Gaffney et al. found that physicians spent about 1.77 hours daily on documentation outside of their assigned hours in the office [Gaffney et al., 2022]. Generation and interpretation of these notes manually can also lead to human errors and medical complications down the road [Bowman, 2013]. At Brigham and Women's Hospital, obstetric anesthesiologists have been experimenting with the use of an AI scribe to summarize patient-doctor conversations; but the workflow has lacked a system to evaluate the quality of generated notes, leading hesitancy for full adoption of LLMs. In such environments where errors are potentially life-threatening, it is important for the caretakers to quickly and accurately flag low-quality or high-risk areas of generated notes on the fly to mitigate risks and streamline the workflow.

Drawing from a related work on evaluating emergency medicine handoff notes, this project focused on evaluating patient notes along the following rubric categories (all score on a 1-5 scale) [Hartman et al., 2024]:

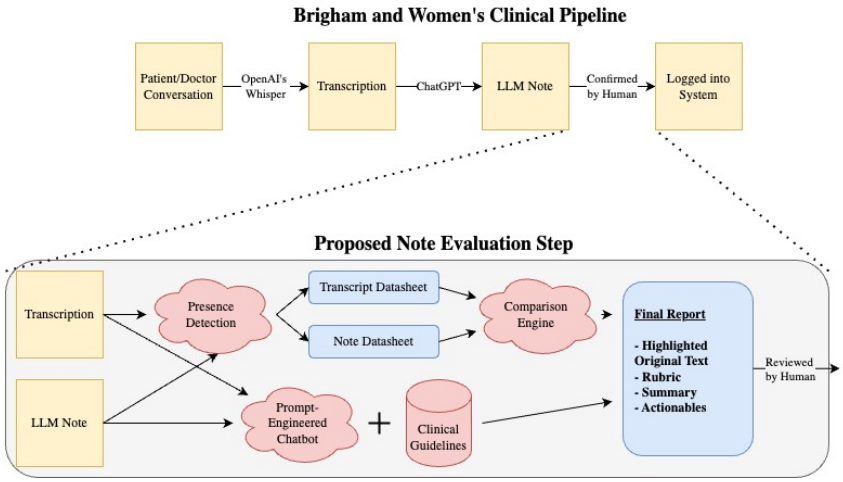


Figure 2: Method Overview

1. Readability: The note uses professional medical language appropriate for an anesthesia consultation.
2. Completeness*: Overall key information included in the summary during the OB patient encounter
3. Curation*: Overall exclusion of irrelevant information.
4. Correctness - Hallucination*: Inventing summary sentences with no basis in the source.
5. Correctness - Knowledge Gap*: Generating statements inconsistent with source.
6. Correctness - Faulty Logic: Inferring logically incorrect summary sentences based on information from the source documents.
7. Correctness - Bias: The note avoids expressing bias toward the patient.
8. Correctness - Overall: Obtained by aggregating all error types, focusing primarily on faulty logic and bias.
9. Patient Safety Risk: The note avoids errors that could directly impact patient care and safety.
10. Usefulness: The note would require minimal corrections before signing and contains accurately captured clinically significant information.

Our proposed method has two parts: a quantitative framework for evaluating correctness, and a qualitative framework for evaluating more nuanced rubric items. All relevant code was published to our open-source repository.

3.2 Rubric-Scoring Method

Discussions with the team at Brigham and Women's Hospital revealed that anesthesiologists were primarily concerned with the following risk factors when evaluating for the factual correctness of an LLM-generated note: Placenta Previa, Placenta Accreta Spectrum, Multiple Gestation, Polyhydramnios, Large for Gestational Age, Fibroids, Preeclampsia, HELLP Syndrome, and Anticoagulation use).

These risk factors are most relevant to the rubric-items above with asterisks next to their titles, as they measure how important information from a conversation is recorded in the corresponding note. For example, "Completeness: Overall key information included in the summary during the OB patient encounter" should measure whether the relevant risk factors for a patient were correctly recorded. In essence, we can think of these rubric items from a classification perspective, allowing us to reformulate them mathematically:

1. Completeness: (No. of correct risk factors in the note) / (No. of risk factors in the transcript)

2. Curation: (No. of risk factors correctly not in the note) / (No. of risk factors not in the transcript)
3. Correctness: Hallucination: (No. of risk factors that are missing in the transcript but exist in the note. Each error deducts a point.)
4. Correctness: Knowledge Gap: (No. of risk factors that are described as positive/negative in the transcript, but not correctly labeled in the note Each error deducts a point.)

We note this approach not only allows us to have more interpretation on the scoring, but it also allows us to present LLMs with the more concrete task of checking for the presence of risk factors, when compared with the task of simply evaluating notes.

Based on these items, our approach proceeds as follows:

1. Use an LLM to automatically label each sentence in both the original transcript and the corresponding LLM-generated note as "Positive", "Negative", or "Missing" for those concepts.
2. Using Table 1, find errors and tabulate them. For instance, if the Transcript does not mention Placenta Previa but the Note says that patient has it, we classify this as a Hallucination Error.
3. Calculate evaluation scores using the aforementioned transformed rubric items.
4. For each error, highlight sentences in the transcript and the sentences in the note that conflict, and present them in the final report as an actionable item.

Transcript:	**Note:**	**Explanation:**
Positive	Positive	No problems
Positive	Negative	Knowledge Gap
Positive	Missing	Knowledge Gap
Negative	Positive	Knowledge Gap
Negative	Negative	No problems
Negative	Missing	Knowledge Gap
Missing	Positive	Hallucination
Missing	Negative	Hallucination
Missing	Missing	No problems

Table 1: Types of Classifications

3.3 RAG-based Method

Qualitative assessments are difficult to transform into classification or other traditional problem formulations, as they rely on greater nuances than simply the presence of concepts. However, we can still improve the performance of vanilla LLMs by improving their knowledge base. We propose using Retrieval-Augmented Generation (RAG) to improve the performance of LLMs by fusing medical data into the model's knowledge base. By integrating this RAG-based LLM with comprehensive analysis of both transcript and AI-generated note, we can identify clinically significant discrepancies and generate targeted recommendations for improvement to increase note quality and potentially clinical outcomes.

We implemented the RAG for our LLM as follows:

1. Construct a knowledge base for obstetric anesthesia using clinical guidelines and medical documentation from Brighman Women's Hospital (eg "Common Terms and Abbreviations Used in Pregnancy Care").
2. Process raw files through AWS's RAG workflow through AWS Knowledge Base. Ingest and chunk manuals into semantically meaningful segments, vectorize them using embedding models, and index them through Amazon OpenSearch Serverless and Pinecone DB.
3. Prompt-engineer for the model to retrieve documents when assessing qualitative dimensions of note quality.

For each evaluated note, our system generated structured actionable improvements using a standardized format:

1. **Issue Type:** Categorized as Safety Risk, Faulty Logic, Bias, Documentation Gap, or Readability Issue
2. **Transcript Evidence:** Direct quote or factual summary from source material
3. **Note Content:** Corresponding content in the generated note or indication of omission
4. **Suggestion:** Specific recommendation for correction

3.4 Assessment Report

Our system produces comprehensive structured evaluation reports for each AI-generated clinical note, offering both quantitative assessment and actionable feedback. The report architecture consists of three integrated components designed to facilitate efficient clinical review:

- **Annotated Clinical Note:** The original AI-generated note with color-coded highlighting that visually identifies problematic content by issue categories such as hallucination, knowledge gap, and faulty logic.
- **Standardized Rubric Assessment:** A detailed evaluation table presenting scores on a 1-5 scale across dimensions. Each score includes a brief justification emphasizing identified issues that are described more extensively in the next section.
- **Structured Actionable Improvements::** A prioritized list of specific recommended modifications with the format mentioned above.

Our system generates an assessment report for each generated note against its original transcript. The report includes the overall score, which is a mixture of qualitative and quantitative metrics previously described, problematic sentences in the generated notes, as well as its issue type and a short justification. For each of the problematic sentences, the report also contains suggestions on how a caretaker might want to modify that part of the note. The anatomy of the report is discussed more in the Results and Discussions portions of the paper.

4 Data and Experiment Setup

We utilized a synthetic dataset of obstetric anesthesia consultations developed Brigham and Women's Hospital for obstetric anesthesiologists. The complete dataset consists of 20 simulated patient-doctor interactions (Patient 1-20). We developed our model with patients 1-5 and 11-15; the rest of the patients were held out for evaluation purposes. All data was securely shared via an institutional Dropbox repository from Brigham and Women's Hospital and contained no protected health information as all cases were synthetic. RAG was implemented using Perplexity, LLM-prompting was conducted with ChatGPT, and other codes were run locally on Python 3.10 in Jupyter Notebook environments.

Each patient in the dataset was generated with the following procedure:

1. Structured skeletons/scripts designed to simulate realistic doctor-patient interactions during anesthesia consultations
2. Simulated consultation conversations performed by medical professionals
3. Automated transcripts generated using OpenAI's Whisper speech-to-text model
4. Clinical notes generated by GPT-4 LLM in-house algorithms
5. Manual scoring by anesthesiologists using a standardized 0-5 scale across multiple quality dimensions

We also utilized comprehensive obstetric anesthesia guidelines and clinical manuals from Brigham and Women's Hospital that served as the knowledge base for our retrieval-augmented generation (RAG) implementation. These resources contained standard protocols, risk assessment frameworks, and best practices for obstetric anesthesia consultations. The LLM models tested were ChatGPT (o4-mini-high from OpenAI), Gemini 2.5 (Google), and Claude 3.7 Sonnet Thinking (Anthropic). However, all results mentioned in this paper used ChatGPT.

We evaluated our two-step method using the following benchmarks

		Ground-Truth Statistics				Results of Automatic Labeling		
Patient #	**Document**	**# of Positive**	**# of Negative**	**# of Missing**	**# Total**	**# Incorrect**	**# Correct**	**% Correct**
6	Transcript	2	5	2	9	2	7	0.7777777777777778
	Note	2	5	2	9	2	7	0.7777777777777778
7	Transcript	4	4	1	9	0	9	1
	Note	4	3	2	9	0	9	1
8	Transcript	4	1	4	9	2	7	0.7777777777777778
	Note	5	2	2	9	2	7	0.7777777777777778
9	Transcript	4	3	2	9	2	7	0.7777777777777778
	Note	4	3	2	9	0	9	1
10	Transcript	3	4	2	9	1	8	0.8888888888888888
	Note	2	3	4	9	3	6	0.6666666666666666
16	Transcript	4	4	1	9	2	7	0.7777777777777778
	Note	4	3	2	9	0	9	1
17	Transcript	2	5	2	9	2	7	0.7777777777777778
	Note	2	5	2	9	1	8	0.8888888888888888
18	Transcript	3	4	2	9	0	9	1
	Note	3	4	2	9	0	9	1
19	Transcript	2	5	2	9	2	7	0.7777777777777778
	Note	2	3	4	9	0	9	1
20	Transcript	2	5	2	9	2	7	0.7777777777777778
	Note	3	4	2	9	2	7	0.7777777777777778
Total	Overall	61	75	44	180	25	155	0.8611111111111112

Table 2: Feature-Wise Statistics on Presence Labeling

1. **Rubric Scoring**: Variability within this method comes from the accuracy of our presence-labeling scheme–in other words, assuming that we label the presence of risk factors correctly, our procedure for classifying errors comes logically from their definitions. Thus, we evaluated this section by comparing automated risk factor labeling with manual. For each patient, we identified the predefined clinical concepts (e.g., medical history, allergies, previous anesthesia experiences) and systematically mapping their presence and representation in both the original transcripts and the AI-generated notes. We then compared these results with our automated concept detection.
2. **RAG**: We evaluated our RAG framework using the same evaluation procedure as MedPalm2. We generated clinical reports with and without RAG, and we asked Dr. Kovacheva to score the quality of the reports, specifically the actionables on the clinical note, under a random and blind study. We then determined the efficacy of RAG through her preferences.

5 Results

Since our rubric scoring method depends heavily on the ability of LLMs to detect the presence of risk factors in sentences, we first ran experiments comparing manual labeling vs LLM labeling. When tested across all patients in our held-out set, our LLMs accurately labeled individual sentences the vast majority of the time (Figure in the Appendix for reference). On a sentence by sentence label, an LLM marking the presence of a risk factor was equivalent to a human reading the note. While this number suggests that our method is generally good at understanding the presence of labels, most sentences do not mention specific risk factors; as a result, the number does not capture how good our model is at discerning positive vs negative presence of concepts. We therefore also compute statistics on a feature-wise level in Table 2. A human being can only have or not have a risk factor; thus, a patient transcript or a patient note should either suggest that a risk factor is present, is not present, or not mention the risk factor at all. This is reflected by the column "# Total" being 9 for all rows. From this perspective, we found that our LLM as Judge - based approach had about a 86% accuracy on a document-by-document level at assessing the presence of features.

We also present some qualitative results in Figures 3, 4, and 5. We can observe that our output found actionable insights based on the risk factors and clinical guidelines. We can also observe that, as a result of some mislabelings of 'positive' or 'negative', our model often produced errors that our manual output didn't. We discuss potential improvements in the Discussion.

Additionally, we also examine the results of our RAG actionables across different iterations of prompt engineering in Figure 3. Base LLM outperformed RAG in both iterations, though RAG improved significantly from 27% to 45% win rate after prompt optimization based on medical expert

Iteration	Approach	Patient ID										
		2	6	7	8	9	10	16	17	18	19	20
1	RAG Score	3	4	4	4	3	3	3	3	3	3	3
	Base LLM Score	4	5	3	3	2	4	4	4	4	4	4
2	RAG Score	3	4	3	4	4	3	4	4	3	3	3
	Base LLM Score	4	3	4	3	3	4	3	3	4	3	4

Table 3: Comparison of RAG and Base LLM Performance Scores

Actionables:

1. **Knowledge gap error: Anticoagulation Use**
   a. Transcript: "And you're just taking the baby aspirin. Is that right? Just baby aspirin every day."
   b. Note: No mention.
   c. Suggestion: Mention Anticoagulation Use in the note.
2. **Hallucination error (highlighted in yellow): HELLP**
   a. Transcript: No mention.
   b. Note: “Medical History: Chronic hypertension, polyhydramnios. No preeclampsia or HELLP.”
   c. Suggestion: Remove incorrect mention of HELLP.
3. **Knowledge gap error: Fibroids**
   a. Transcript: "You don't have any fibroids currently, is that correct? No, not that I know of."
   b. Note: No mention.
   c. Suggestion: Mention that the patient does not have Fibroids.
4. **Hallucination error (highlighted in green): Placenta Accreta**
   a. Transcript: No mention.
   b. Note: “No history of uterine rupture, uterine inversion, or placenta accreta.”
   c. Suggestion: Remove incorrect mention of HELLP.

Figure 3: Example Output of Rubric-Based Method

scoring feedback. We believe that these results are likely attributable to our implementation of RAG, particularly how we computed chunks of our documents and our prompt. Further iterations could help test different prompt engineering approaches to continue enhancing RAG performance for clinical documentation.

6 Discussion

Our study demonstrates that the developed evaluation framework for AI-generated obstetric anesthesia consultation notes effectively addresses a critical clinical documentation challenge. The LLM-based evaluation approach, coupled with our structured reporting system, shows promise as both a quality assessment tool and an intelligent review assistant that significantly reduces clinician review time.

Output 2:			**Score (1-5): 3**
Issue Type	**Transcript**	**Note**	**Suggestion**
Documentation Gap	Now and sometimes in these events we need to give blood transfusions. Are you okay receiving blood only if necessary? - "Yeah, my doctor mentioned that this can happen."	Not mentioned	Document patient's consent for blood transfusion if necessary in the Assessment and Plan section.
Faulty Logic	No explicit discussion of absence of bleeding episodes in current pregnancy	She has not had any bleeding episodes in this pregnancy.	Remove this statement or rephrase as "No bleeding episodes in current pregnancy reported during consultation."
Documentation Gap	And this is pregnancy, did you utilize IVF for this pregnancy? - "No."	Not explicitly mentioned	Add "Non-IVF pregnancy" to the obstetric history section for completeness.
Faulty Logic	I started taking Lovenox during the pregnancy because I previously had a clot.	She is on prophylactic Lovenox due to a history of deep vein thrombosis (DVT)	Clarify whether Lovenox is being used prophylactically or therapeutically, as this impacts neuraxial anesthesia timing decisions.

Figure 4: Example Output of RAG-Based Method

	Patient 6	Patient 7	Patient 8	Patient 9
Manual Output	- Fibroids has a knowledge gap error - Multiple Gestation has a hallucination error	- Anticoagulation use has a knowledge gap error - Preeclampsia has a knowledge gap error	- Preeclampsia has a hallucination error **- Multiple Gestation has a hallucination error**	- Polyhydramnios has a knowledge gap error **- Anticoagulation use has a hallucination error**
Automatic Output	- Polyhydramnios has a hallucination error **- HELLP has a hallucination error** **- Large for Gestational Age has a hallucination error** - Multiple Gestation has a hallucination error	- Preeclampsia has a knowledge gap error - Anticoagulation use has a knowledge gap error	- Preeclampsia has a hallucination error **- HELLP has a hallucination error**	- Polyhydramnios has a knowledge gap error **- HELLP has a knowledge gap error**
	Patient 10	**Patient 16**	**Patient 17**	**Patient 18**
Manual Output	**- The transcript says that the presence of Polyhydramnios is Negative** **- Large for Gestational Age has a knowledge gap error** **- HELLP has a knowledge gap error** - Anticoagulation use has a hallucination error	- Multiple Gestation has a hallucination error **- HELLP has a knowledge gap error** - Preeclampsia has a knowledge gap error	- Fibroids has a knowledge gap error - Multiple Gestation has a hallucination error	- Preeclampsia has a knowledge gap error - Multiple Gestation has a hallucination error
Automatic Output	- Anticoagulation use has a hallucination error **- Placenta Previa has a hallucination error**	- Preeclampsia has a knowledge gap error **- Large for Gestational Age has a hallucination error** - Multiple Gestation has a hallucination error	- Fibroids has a knowledge gap error - Multiple Gestation has a hallucination error **- Polyhydramnios has a hallucination error**	- Preeclampsia has a knowledge gap error - Multiple Gestation has a hallucination error
	Patient 19	**Patient 20**		
Manual Output	**- Preeclampsia has a knowledge gap error** - Multiple Gestation has a hallucination error	- Multiple Gestation has a hallucination error - Polyhydramnios has a knowledge gap error **- Anticoagulation use has a knowledge gap error** **- Large for Gestational Age has a knowledge gap error**		
Automatic Output	- Multiple Gestation has a hallucination error **- Anticoagulation use has a knowledge gap error**	- Multiple Gestation has a hallucination error - Polyhydramnios has a hallucination error **- Preeclampsia has a knowledge gap error** **- Placenta Previa has a knowledge gap error** **- Fibroids has a knowledge gap error**		

Figure 5: Comparison of errors caught by manual annotation versus automated pipeline (discrepancies are bolded)

To integrate into clinical practice, further development is needed to unify the currently separate components (RAG-LLM scoring and concept labeling detection) into a cohesive automated pipeline. A user-friendly front-end interface would enhance usability through intuitive presentation of findings and actionable recommendations for clinicians. Designing this interface to align with existing clinical documentation systems would minimize workflow disruption during implementation. Clinical integration would require thorough testing with existing systems at Brigham and Women's Hospital to ensure compatibility with current documentation practices and clinical workflows.

While our implementation shows promising results, several limitations warrant consideration. The use of synthetic data, while necessary for development, may not fully capture the nuances of real patient-provider interactions, including variations in complications and patient info. The professional medical actors used in simulations likely produced more structured and medically precise conversations than might occur with actual patients who possess varying levels of health literacy and communication styles.

Additional limitations include the reliance on input from a single medical expert (Dr. Kovacheva) due to time constraints, potentially limiting generalization across different clinical contexts and provider preferences. Though expert feedback suggests promising workflow efficiency improvements, we could not directly measure time savings in actual clinical practice. Our efficiency assumptions derive from structured feedback rather than quantitative measurements in clinical environments.

Finally, our current implementation may be modified to consider clinicians' priors. We did consider this element originally during our method design. However, when speaking with clinicians originally, we believed that the use of 'priors' was inappropriate for this task, as doctors prefer to have all assumptions spoken and documented explicitly than rely on shared understanding. However, while evaluating our documents, we observed a large number of hallucination errors, as a result of these unspoken priors (for instance, pregnancies, without the mention of multiple gestation, are assumed to be singleton). The addition of such assumptions, along with the improvement of the RAG mentioned in our results section, would likely improve the quality of our generated actionables.

Future research should explore system performance with real patient consultations, incorporate diverse clinical expert perspectives, and quantify efficiency gains through implementation studies. Expanding the range of tested clinical scenarios would strengthen the system's robustness for widespread deployment. Despite these limitations, the potential for this evaluation framework appears considerable, particularly as healthcare systems increasingly adopt LLMs and generative AI in clinical documentation workflows.

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Davin Jeong and Waris Tuchinda conducted the initial comparison between the LLM and BERT-based scoring engines. Bolanle Oladeji and Waris Tuchinda performed manual labeling of the original transcripts and corresponding LLM-generated notes. Davin Jeong developed the rubric-scoring method, which uses presence labeling to find quantitative scores and actionable insights. Dat Tran developed the qualitative scoring framework and evaluation using the RAG-enhanced LLM. All team members contributed actively to progress meetings, ideation, and the preparation of the final paper.

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8 Appendix

This appendix contains supplementary materials for our research. All prompts, documents for RAG, and associated code can be found in our GitHub repository:

https://github.com/huskydj1/ML4HC_RubricScoring

The repository includes:

- Prompt templates for presence labeling
- Prompt configurations for RAG assessments
- Example synthetic data used evaluation pipeline
- Implementation code and evaluation scripts

We encourage interested readers to explore the repository for a comprehensive understanding of our methodology and to facilitate reproducibility of our results.

8.1 Additional Figures

		Ground-Truth Statistics				Results of Automatic Labelling		
Patient #	**Document**	**# of Positive**	**# of Negative**	**# of Missing**	**# Total**	**# Incorrect**	**# Correct**	**% Correct**
6	Transcript	14	8	2075	2097	11	2086	0.994754411063424
	Note	8	6	526	540	6	534	0.9888888888888889
7	Transcript	12	10	2354	2376	2	2374	0.9991582491582491
	Note	15	5	592	612	4	608	0.9934640522875817
8	Transcript	8	1	2115	2124	7	2117	0.9967043314500942
	Note	12	3	516	531	5	526	0.9905838041431262
9	Transcript	10	3	1364	1377	4	1373	0.9970951343500363
	Note	17	3	574	594	2	592	0.9966329966329966
10	Transcript	11	5	1577	1593	3	1590	0.9981167608286252
	Note	6	4	548	558	7	551	0.9874551971326165
16	Transcript	16	11	3330	3357	6	3351	0.998212689901698
	Note	8	4	483	495	8	487	0.9838383838383838
17	Transcript	6	11	1684	1701	9	1692	0.9947089947089947
	Note	5	5	521	531	9	522	0.9830508474576272
18	Transcript	4	9	1850	1863	0	1863	1
	Note	9	5	364	378	13	365	0.9656084656084656
19	Transcript	4	7	1051	1062	3	1059	0.9971751412429378
	Note	5	3	370	378	7	371	0.9814814814814815
20	Transcript	4	8	1833	1845	2	1843	0.9989159891598915
	Note	5	4	567	576	2	574	0.9965277777777778
Total	Overall	179	115	24294	24588	110	24478	0.9955262729786888

Table 4: Sentence-Wise Statistics on Presence Labeling